

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts

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ABSTRACT

Model Predictive Control (MPC) has served as the industrial benchmark for constrained multivariable control across process operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse unstructured natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract unit tests, and preemptive trip bounds. High-level cognitive directives are translated at runtime into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across parallel trajectory rollouts via GPU/WebGPU compute shaders, augmented by an online recursive parameter observer.

We evaluate the physical dynamics on an exothermic Continuous Stirred-Tank Reactor (CSTR) undergoing non-linear kinetic surges across a fully reproducible benchmark suite spanning 20 randomized seeds with varying surge magnitudes ($C_{A0} \in [+10\%, +30\%]$, $T_0 \in [+5\text{ K}, +12\text{ K}]$, and $UA \in [-10\%, -35\%]$) and advance-warning advisory lead time ($t_{\text{lead}} = 3.5\text{ s}$). Numerical dynamic optimization demonstrates that with zero advance warning, the combined jacket transport delay ($\tau_j = 2.0\text{ s}$) and valve slew rate limit (12 K/s) physically prevent any purely reactive controller from avoiding thermal runaway ($T_{\text{max}} \geq 440\text{ K}$), establishing a physical floor of 6/20 trips across all feedback controllers (Linear MPC: 6/20 trips, 9.49 K RMSE; NMPC: 6/20 trips, 9.62 K RMSE; MPPI: 6/20 trips, 9.79 K RMSE).

When advance preview lookahead ($t_{\text{lead}} = 3.5$ s) is supplied, trips drop to 1/20 across all preview controllers, with failure occurring exclusively on Seed 18 (an extreme corner realization requiring >4.5 s of lead time). Crucially, our ablation isolates that the supervisory layer's quantitative advantage (0.95 ± 0.40 K non-trip RMSE vs. 1.81 ± 1.11 K for standard preview MPPI) stems from coordinating an anticipatory pre-cooling ramp that avoids **reaction quenching**: naive rule-based step switching quenches the reactor, accumulates unreacted feed, and triggers delayed thermal blowout (8/20 trips, 9.44 K RMSE). We present the full system architecture, formal Ring 0 contract invariants, and offline prompt evaluation protocols, establishing a rigorous foundation for agent-directed supervisory process control.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on millisecond timescales via convex quadratic programming (QP), linear approximations degrade on highly non-linear chemical plants governed by exponential kinetics, such as Continuous Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, purely reactive controllers exhibit severe tracking error, control chattering, and valve saturation.

1.1 Related Work: Advanced & Learning-Based MPC

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018). Physics-informed neural network formulations (Raissi et al., 2019) have inspired domain-constrained residual observers that embed conservation laws into state estimation.
- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU compute shaders.

1.2 The Role of the Supervisory Layer: Lookahead vs. Reasoning

A critical insight in receding-horizon control is that numerical solvers optimize over a fixed finite horizon H_p (e.g., 4.0 s). When future disturbances are known in advance, mathematical solvers with preview can naturally pre-actuate within their horizon. However, in actual plant operations, forecasts arrive as human shift handover notes, laboratory feed analysis reports, or upstream DCS alarms. Classical MPC cannot parse these qualitative inputs.

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner interprets operator directives and contextual operational runbooks, translates them into quantitative cost manifolds, and verifies them through deterministic contract tests before any parameter update reaches physical actuators.

2. The Agentic MPC Architecture

2.1 The Two Halves: The Knower and The Doer

- **The Knower (Neural Reasoner):** Operates in *Intelligence Space*, translating operator directives into cost manifold parameters $\theta_{\mathcal{M}} = \{q_{\text{temp}}, q_{\text{barrier}}, u_{\text{target}}, \lambda\}$. Inference runs on dedicated cloud LPUs or server GPUs (e.g., Qwen-2.5-32B), with lightweight client-side WebGPU shaders (e.g., Qwen-2.5-0.5B via @mlc-ai/web-llm) available for offline edge operation.
- **The Doer (Executive Function Harness):** A deterministic verification kernel enforcing Ring 0 invariants (actuator saturation and slew rate limits), running automated contract unit tests, and operating preemptively below plant emergency shutdown thresholds.

2.2 The Four Structured Memory Subsystems

Memory Store	Storage Modality	Functional Role	CSTR Realization
 The Playbook	Procedural (SKILL.md)	Verified operational runbooks with deterministic pre/post-conditions	cstr_runaway_mitigation.md verified via automated contract tests
 The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0\text{ K}$, nominal $T_{\text{ref}} = 350.0\text{ K}$
 The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: V-101 → J-101 → CV-201 → M-101 → CH-3

Memory Store	Storage Modality	Functional Role	CSTR Realization
 The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (MUTEX_PREC00L) preventing conflicting concurrent directives

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline (Jacobian QP)

The linear discrete-time optimal control problem is solved as a condensed Quadratic Program (QP) around the textbook steady state ($C_{A,\text{ref}} = 0.50 \text{ mol/L}$, $T_{\text{ref}} = 350.0 \text{ K}$, $T_{c,\text{base}} = 300.0 \text{ K}$):

$$\min_U \frac{1}{2} U^T H_{\text{qp}} U + g^T U \quad \text{subject to: } u_{\min} \leq u_k \leq u_{\max}, \quad |\Delta u_k| \leq \Delta u_{\max}$$

The continuous Jacobian matrix A has eigenvalues $[-0.0076, +0.0472] \text{ s}^{-1}$, exhibiting an open-loop thermal runaway pole ($\tau_{\text{growth}} \approx 21.2 \text{ s}$). State penalty is $Q = \text{diag}(10.0, q_T)$, $R = 0.02$, input bounds $[280, 360] \text{ K}$, and slew limit 12 K/s .

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI evaluates $K = 1,024$ stochastic control rollouts sampled from $\mathcal{N}(0, \Sigma)$ with covariance $\Sigma = 6.0^2 I$ and temperature $\lambda = 10.0$. Discretized at rollout step $\Delta t_{\text{ctrl}} = 0.2 \text{ s}$:

$$x_{k+1} = f(x_k, v_k)$$

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (q_{\text{temp}}(T_k - T_{\text{ref}})^2 + q_{\text{barrier}} \max(0, T_k - T_{\text{barrier}})^2) + \frac{\lambda}{2} \sum_{k=0}^{H_p-1} \epsilon_k^T \Sigma^{-1} \epsilon_k$$

The optimal control update is computed via softmax importance weighting:

$$u_t^* = u_t + \sum_{m=1}^K w(U^{(m)}) \epsilon_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp(-\frac{1}{\lambda}(S(U^{(m)}) - \min_j S(U^{(j)})))}{\sum_{k=1}^K \exp(-\frac{1}{\lambda}(S(U^{(k)}) - \min_j S(U^{(j)})))}$$

4. Benchmark Case Study: Exothermic CSTR

4.1 Numerical Dynamic Oracle Feasibility

Dynamic trajectory optimization over the benchmark kinetic surge demonstrates the physical controllability limits under actuator saturation ($T_c \geq 280 \text{ K}$, $|\dot{T}_c| \leq 12 \text{ K/s}$):

Pre-Cooling Lead Time t_{lead}	Peak Reactor Temp T_{max} (K)	Emergency SIS Trip (≥ 385.0 K)	Operational Controllability Outcome
0.0 s (Reaction at Onset)	444.9 K	TRIP (Breached)	Unsurvivable runaway due to jacket lag ($\tau_j = 2$ s) & slew cap
1.0 s Lead Time	443.4 K	TRIP (Breached)	Insufficient thermal extraction ahead of exponential surge
2.0 s Lead Time	378.2 K	SAFE (Zero Trip)	Modulated dynamic pre-cooling contains surge below trip
2.5 s Lead Time	367.8 K	SAFE (Zero Trip)	Stable containment with 17.2 K safety margin
3.0 s Lead Time	361.2 K	SAFE (Zero Trip)	Smooth containment
3.5 s Lead Time	356.5 K	SAFE (Zero Trip)	Robust containment (28.5 K margin to trip)
5.0 s Lead Time	350.0 K	SAFE (Zero Trip)	Fully absorbed thermal surge clamped at nominal setpoint

5. Verification & Safety Architecture

Agentic MPC enforces defense-in-depth: Ring 0 invariant checking ($q_T \geq 0.1$, $q_{\text{barrier}} \geq 1000$, $u \in [280, 360]$ K, $\text{slew} \leq 12$ K/s), automated contract unit tests on the active fouled plant model (<30 ms), a software preemptive breaker at 382 K, and an independent hardware SIS scram at 385 K (failsafe cooling valve full open, feed shut off).

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Benchmark Across 20 Randomized Seeds

Table 2 presents authentic simulation results across 20 randomized seeds computed directly from discrete QP, L-BFGS-B NMPC, and GPU MPPI path integral solvers:

Configuration	Controller Type	Non-Trip RMSE (K)	Overall RMSE (K)	Peak T (K)	Slew (K/s)	SIS Trips	Settle
Baseline 1 (qT=1)	Linear MPC ($q_T = 1.0$, No Forecast)	2.56 ± 1.36	16.26 ± 5.52	389.3 ± 14.2	3.29 ± 0.95	12 / 20 (60%)	8 / 20 (9.8 s)

Configuration	Controller Type	Non-Trip RMSE (K)	Overall RMSE (K)	Peak T (K)	Slew (K/s)	SIS Trips	Settle
Baseline 1 (qT=10)	Linear MPC ($q_T = 10.0$, No Forecast)	2.45 ± 1.41	9.49 ± 5.34	372.1 ± 13.5	4.85 ± 1.12	6 / 20 (30%)	13 / 20 (7.1 s)
Baseline 2a	NMPC (L-BFGS-B, No Forecast)	2.48 ± 1.47	9.62 ± 5.37	372.9 ± 13.7	5.04 ± 1.29	6 / 20 (30%)	13 / 20 (6.9 s)
Ablation 1	MPPI (Static θ^* , No Forecast)	2.98 ± 1.75	12.03 ± 5.57	378.9 ± 14.6	2.44 ± 0.38	8 / 20 (40%)	11 / 20 (8.6 s)
Ablation 2	MPPI + Observer (No Forecast)	2.60 ± 1.28	9.79 ± 5.39	373.4 ± 13.9	2.42 ± 0.38	6 / 20 (30%)	14 / 20 (11.6 s)
Baseline 2b	NMPC (With Forecast Preview)	1.47 ± 0.85	2.82 ± 2.93	356.7 ± 7.6	6.41 ± 1.03	1 / 20 (5%)	19 / 20 (2.6 s)
Ablation 3a	MPPI + Observer (With Forecast)	1.81 ± 1.11	3.11 ± 2.99	357.3 ± 7.7	2.72 ± 0.19	1 / 20 (5%)	18 / 20 (5.7 s)
Ablation 3b	Rule Supervisor + MPPI	2.50 ± 1.66	9.44 ± 4.92	381.1 ± 17.1	3.77 ± 0.37	8 / 20 (40%)	12 / 20 (13.1 s)
Full System	Agentic MPC (Full Architecture)	0.95 ± 0.40	2.17 ± 2.57	355.5 ± 6.2	3.36 ± 0.19	1 / 20 (5%)	19 / 20 (2.9 s)

6.2 Deconstructing the Supervisory Advantage

Ablation across the 19 non-trip seeds isolates the quantitative mechanism behind the supervisory schedule:

Configuration	Non-Trip RMSE (K)	Physical Mechanism
MPPI + Forecast Preview	1.81 K	Unconstrained optimal preview tracking
+ Ramp Cap Only	1.12 K	Modulates pre-cooling to 282 K, preventing reaction quenching

Configuration	Non-Trip RMSE (K)	Physical Mechanism
+ Elevated Barrier Weights Only	1.36 K	Stiff penalty keeps state strictly within safe envelope
Both (Anticipatory Supervisory Schedule)	0.95 K	Coordinated pre-cooling containment (+0.86 K gain, $p = 0.0088$)

7. Industrial Deployment Limits & Conclusion

While Agentic MPC demonstrates robust supervisory capabilities in simulation, certified industrial deployment requires strict separation of concerns. Under IEC 61511 / ISA-84, safety instrumented functions must remain physically independent, deterministic, and SIL-rated. The agentic layer operates solely in the basic process control system (BPCS) supervisory space and must never share hardware, sensors, or actuators with the emergency shutdown interlock.

Systematic ablation across eight control configurations demonstrates that while numerical preview lookahead provides the essential physical mechanism of thermal stabilization, the supervisory agentic layer provides critical directive translation, formal contract gating, and smooth pre-cooling modulation that eliminates the hazardous quenching trips of naive rule-based switching. Automated contract testing and Ring 0 invariant verification provide the necessary deterministic gating to transition agentic control systems toward industrial deployment.

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